



Retraining-free Merging of Sparse MoE via Hierarchical Clustering

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Abstract

To address Sparse Mixture-of-Experts (some) models' extensive memory requirements of expert components, this paper introduces **Hierarchical Clustering for Sparsely activated Mixture of Experts (HC-SMoE)**, a task-agnostic expert merging framework for parameter reduction without retraining. HC-SMoE introduces a novel hierarchical clustering approach based on expert outputs to ensure merging robustness independent of routing decisions. We evaluate our method across multiple zero-shot language tasks on state-of-the-art models including Qwen and Mixtral. The experimental results validate HC-SMoE's superior performance and practical applicability for real-world deployments.

Background

Sparsely Activated Mixture-of-Experts (SMoE)

Consider an input token x , a set of expert neural networks $\{E_1, E_2, \dots, E_n\}$, and a router network R . The output y of an SMoE layer can be expressed as:

$$y = \sum_{i=1}^n P_i(x) \cdot E_i(x) = \sum_{i=1}^n \text{softmax}(\text{topK}(R(x))) \cdot E_i(x). \quad (1)$$

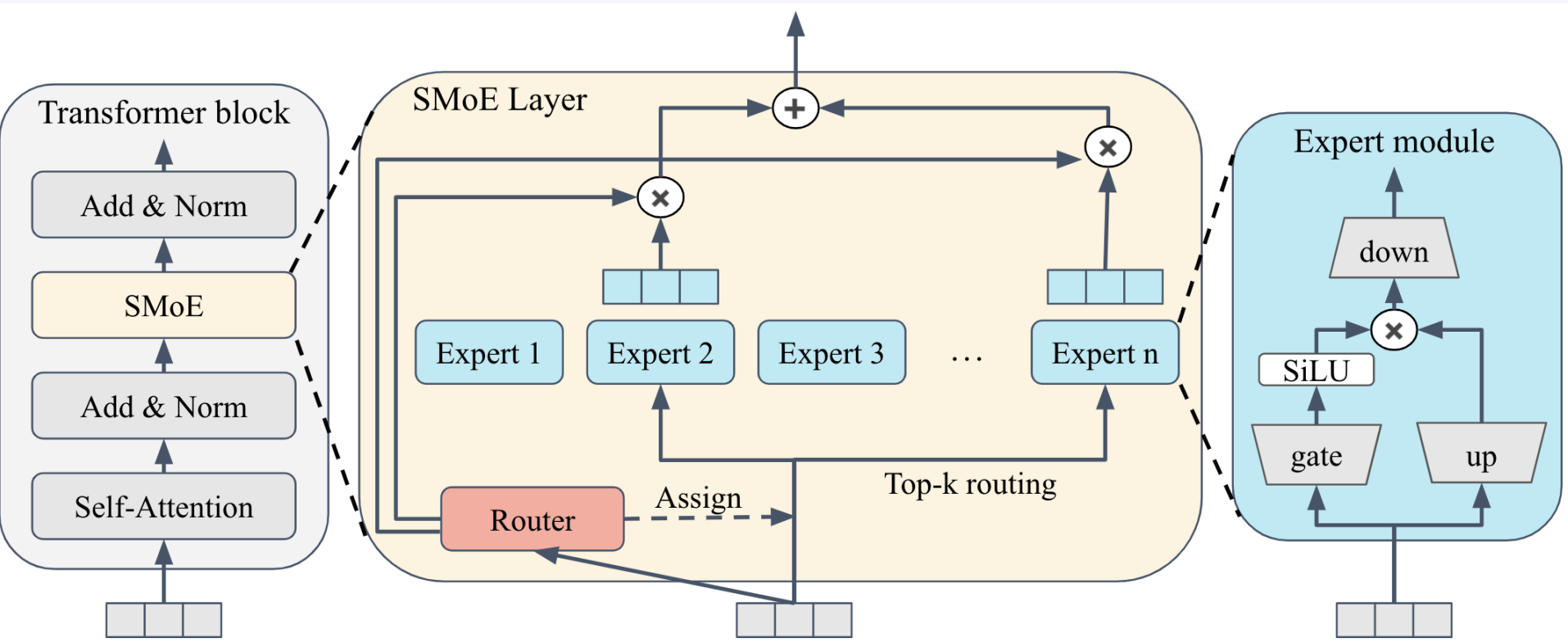


Figure 1. General architecture of SMoE.

Model Compression Techniques for SMoE

Pruning-based Methods

TSEP [1] reduces active experts through iterative fine-tuning for specific downstream tasks. S-prune [2] accumulates global router-scores $P(x)$ and retains top-scoring experts. O-prune [3] determines expert retention counts per layer and evaluates all possible expert combinations to select configurations that minimize output deviation from the original model.

$$\min_C ||\mathcal{F}'(x, C) - \mathcal{F}(x)||_F \text{ s.t. } C \subseteq \{E_1, \dots, E_n\}, |C| = r. \quad (2)$$

However, these approaches discard potential knowledge from pruned experts as well as their corresponding router parts.

Merging-based Methods

M-SMoE [4] first selects dominant experts based on activation frequency to decide which experts to retain in each layer, then uses router logits to group experts, followed by frequency-based merging. Nevertheless, this approach faces three main issues: (1) frequency is an unreliable indicator for deciding number of experts, (2) overlooking high-frequency experts' functional similarities, and (3) grouping based on router logits.

Method	Retraining-free	Task-agnostic	Scalable	Strategy
TSEP	✓	✓	✓	Pruning
O-prune	✓	✓	✗	Pruning
S-prune	✓	✓	✓	Pruning
F-prune	✓	✓	✓	Pruning
M-SMoE	✗	✗	✓	Merging
HC-SMoE	✓	✓	✓	Merging

Table 1. A Comparison of different approaches for reducing the number of experts in SMoE.

Theoretical Justification

Clustering Optimality and Approximation Bound

Let $\{G_j\}_{j=1}^r$ denote a partition sets of the expert indices into r clusters. We define the average-merged expert for group j as Eq.3, where $g(i)$ is the function that maps expert E_i to its cluster index $j = g(i)$. Then HC-SMoE output can be expressed as Eq. (4).

$$\bar{E}_j(x) := \frac{1}{|G_j|} \sum_{i \in G_j} E_i(x). \quad (3) \quad y_{HC}(x) = \sum_{i=1}^n P_i(x) \cdot \bar{E}_{g(i)}(x). \quad (4)$$

The approximation error between the original MoE output and the HC-SMoE output is formulated as:

$$||y_{orig}(x) - y_{HC}(x)||^2 = ||\sum_{i=1}^n P_i(x)(E_i(x) - \bar{E}_{g(i)}(x))||^2 \leq \sum_{i=1}^n P_i(x) \cdot ||E_i(x) - \bar{E}_{g(i)}(x)||^2, \quad (5)$$

by Jensen's inequality applied to the convex function $||\cdot||^2$. This inequality demonstrates that the HC-SMoE approximation error is bounded by the weighted intra-cluster variance. To reduce the approximation error $||y_{orig}(x) - y_{HC}(x)||^2$, we opt to to minimize the intra-cluster variance $||E_i(x) - \bar{E}_{g(i)}(x)||^2$.

Finding optimal solution, defined as $OPT := \min ||E - \bar{E}||^2$, is an NP-hard problem. Hierarchical clustering with average linkage is a polynomial-time algorithm that provides an approximation to the optimal clustering. In the worst case, it guarantees a solution satisfying $||E - \bar{E}||^2 \leq 3 \cdot OPT$ [5]. Therefore, we adopt this computationally efficient method in place of the naive exhaustive approach.

Methodology

Hierarchical Clustering for Sparsely Activated Mixture of Experts (HC-SMoE)

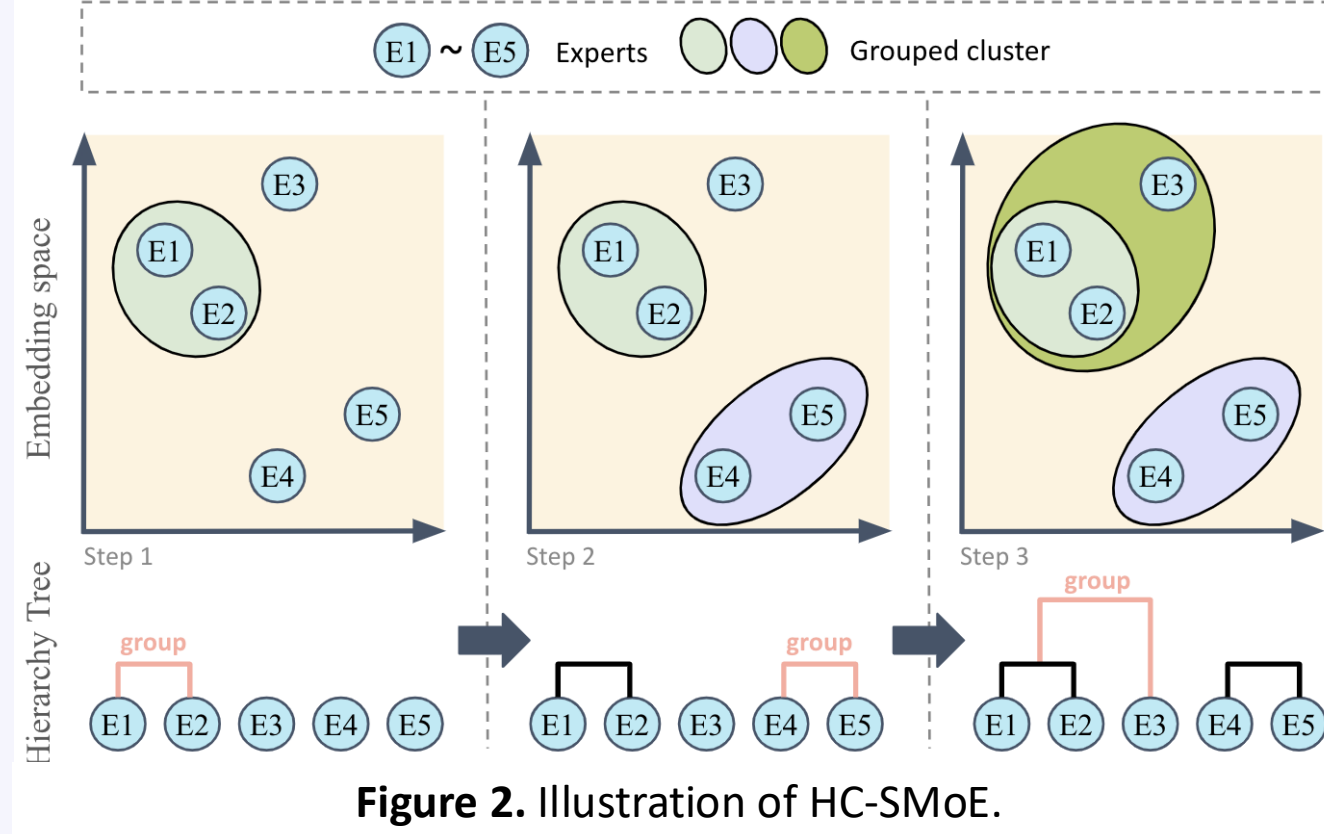


Figure 2. Illustration of HC-SMoE.

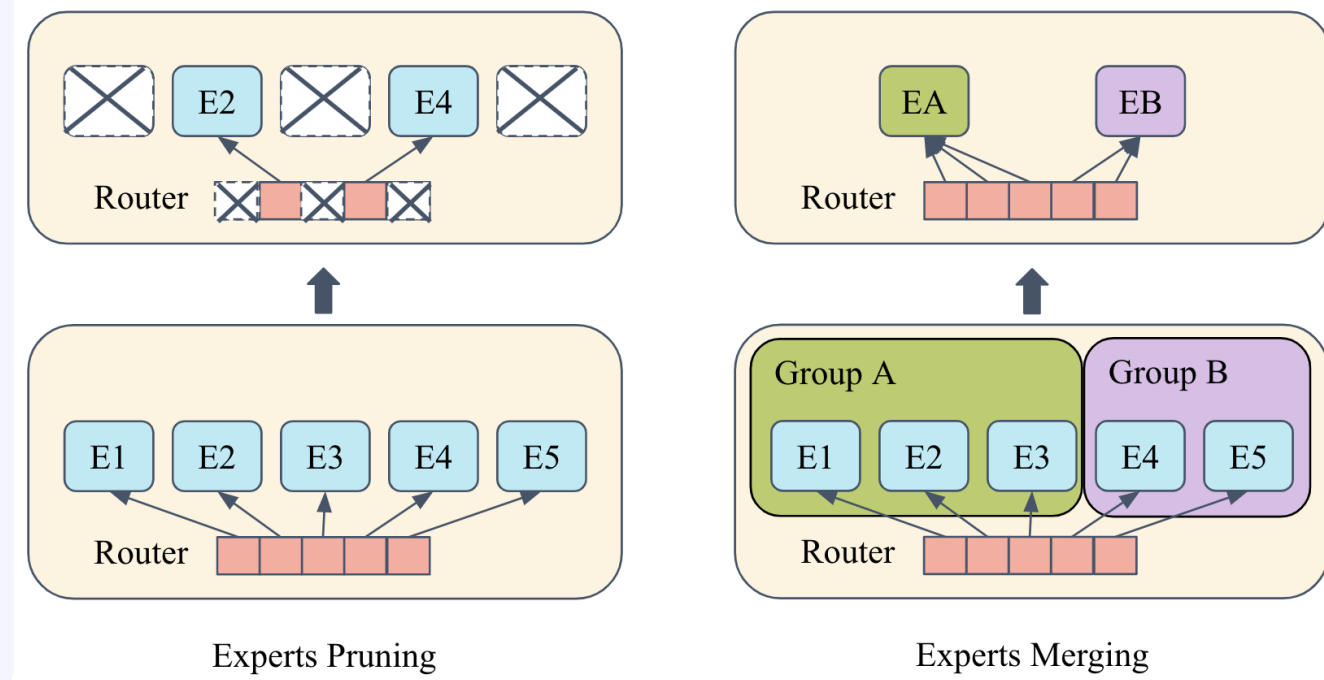


Figure 3. Comparison of expert pruning and merging strategies.

HC-SMoE is a *retraining-free*, *scalable*, and *task-agnostic* framework. HC combines experts through a bottom-up agglomerative process: starting with each expert as a singleton cluster, it recursively combines the most functionally similar pairs while continuously recalculating inter-cluster distances. HC circumvents trapping clusters in suboptimal local minima through its dendrogram-based structure, which accommodates heterogeneous cluster shapes and sizes without requiring explicit assumptions. The clustering process requires two components: (1) a distance metric for measuring differences between expert output vectors, and (2) a linkage strategy for determining inter-cluster distances. We choose Euclidean distance and average linkage on expert-output metric as our method.

Cluter	Metric	ARC-c	BoolQ	OBQA	RTE	Average
K-fix	<i>rl</i>	0.2031	0.4015	0.162	0.4838	0.3126
	<i>weight</i>	0.2073	0.4960	0.166	0.509	0.3446
	<i>eo</i>	0.2184	0.3786	0.148	0.5343	0.3198
	<i>eo</i>	0.2014	0.4168	0.142	0.5018	0.3155
K-rnd	<i>weight</i>	0.2108	0.533	0.174	0.5379	0.3639
	<i>eo</i>	0.3370	0.6398	0.224	0.6065	0.4518
	<i>eo</i>	0.3532	0.7535	0.228	0.6606	0.4988

Table 3. Comparison between the proposed HC-SMoE and K-means clustering approaches on Qwen 30x2.7B.

Linkage	Metric	ARC-c	BoolQ	OBQA	RTE	Average
None	None	0.3951	0.8135	0.310	0.7329	0.5629
Single	<i>rl</i>	0.2398	0.3792	0.180	0.5054	0.3261
	<i>weight</i>	0.3695	0.7676	0.254	0.7004	0.5229
	<i>eo</i>	0.3592	0.7942	0.270	0.7292	0.5382
	<i>eo</i>	0.3677	0.7694	0.248	0.7329	0.5295
Complete	<i>weight</i>	0.2363	0.4446	0.178	0.6137	0.3682
	<i>eo</i>	0.2338	0.6037	0.210	0.5162	0.3909
	<i>eo</i>	0.2073	0.3801	0.172	0.5018	0.3153
Average	<i>weight</i>	0.3788	0.7645	0.250	0.7004	0.5234
	<i>eo</i>	0.3660	0.7948	0.272	0.7509	0.5459

Table 2. Different linkage method and grouping metric comparisons of hierarchical clustering on Qwen 45x2.7B.

Experiments

Subsection Title

We conduct experiments on two SMoE models: Qwen1.5-MoE-A2.7B (henceforth Qwen) [6] and Mixtral 8x7B [7]. For our comparisons, three pruning baselines are employed: O-prune [3], S-prune [2] and F-prune. HC-SMoE demonstrates consistent superiority over these baselines, irrespective of model size, and proves applicable to different numbers of SMoE experts.

Model	Method	ARC-c	ARC-e	BoolQ	HellaSwag	MMLU	OBQA	RTE	Winogrande	Average
Qwen 60x2.7B	None	0.3951	0.7012	0.8135	0.5932	0.6047	0.310	0.7329	0.6559	0.6008
	O-prune (10^5)	0.3268	0.6111	0.7566	0.5388	0.5150	0.268	0.6498	0.6330	0.5374
	F-prune	0.3490	0.5989	0.7618	0.5441	0.4560	0.282	0.7690	0.6409	0.5502
	S-prune	0.3464	0.6061	0.7128	0.5228	0.4930	0.264	0.6534	0.5935	0.5240
	M-SMoE	0.3473	0.6157	0.7544	0.5157	0.4182	0.262	0.7292	0.6377	0.5350
Qwen 45x2.7B	HC-SMoE (avg)	0.3660	0.6578	0.7948	0.5520	0.5332	0.272	0.7509	0.6464	0.5716
	HC-SMoE (single)	0.3592	0.6578	0.7942	0.5578	0.5360	0.270	0.7292	0.6472	0.5689
	O-prune (10^5)	0.2568	0.4449	0.6496	0.4351	0.2907	0.202	0.6065	0.5375	0.4279
	F-prune	0.2765	0.4718	0.6587	0.4330	0.3023	0.230	0.6570	0.5927	0.4528
	S-prune	0.2500	0.4756	0.6388	0.4041	0.3471	0.196	0.6209	0.5146	0.4309
Qwen 30x2.7B	M-SMoE	0.1945	0.2786	0.4462	0.2837	0.2475	0.160	0.4477	0.5185	0.3221
	HC-SMoE (avg)	0.3532	0.6149	0.7535	0.4695	0.4534	0.228	0.6606	0.6456	0.5223
	HC-SMoE (single)	0.3524	0.6153	0.7661	0.4661	0.4537	0.228	0.6534	0.6306	0.5207

Table 2. Zero-shot comparison of Qwen1.5-MoE-A2.7B-Chat.

Model	Method	ARC-c	ARC-e	BoolQ	HellaSwag	MMLU	OBQA	RTE	Winogrande	Average
Mixtral 8x7B	None	0.5648	0.8422	0.8505	0.6490	0.6712	0.350	0.7112	0.7593	0.6748
	O-prune	0.5205	0.8009	0.8352	0.6115	0.5741	0.316	0.6606	0.7719	0.6363
	F-prune	0.5009	0.7904	0.7725	0.5990	0.5099	0.326	0.5596	0.7672	0.6032
	S-prune	0.4991	0.7891	0.7801	0.5984	0.5103	0.340	0.5704	0.7735	0.6076
	M-SMoE	0.2619	0.5564	0.5208	0.4320	0.2503	0.194	0.5271	0.5848	0.4159
Mixtral 6x7B	HC-SMoE (avg)	0.5145	0.8043	0.8554	0.6142	0.6043	0.324	0.6715	0.7514	0.6425
	HC-SMoE (single)	0.5154	0.8123	0.8554	0.6163	0.6053	0.310	0.6715	0.7403	0.6408
	O-prune	0.4394	0.7327	0.8046	0.5660	0.4584	0.286	0.5668	0.7285	0.5728
	F-prune	0.4352	0.7290	0.7520	0.5293	0.3739	0.290	0.5560	0.7245	0.5487
	S-prune	0.2235	0.4339	0.6300	0.4250	0.2554	0.188	0.5235	0.5699	0.4062
Mixtral 4x7B	M-SMoE	0.2216	0.2765	0.4954	0.2767	0.2452	0.108	0.4910	0.4964	0.3251
	HC-SMoE (avg)	0.4573	0.7454	0.8018	0.5709	0.4571	0.270	0.5523	0.7285	0.5729
	HC-SMoE (single)	0.4642	0.7483	0.8321	0.5781	0.4895	0.280	0.5884	0.7206	0.5877

Table 3. Zero-shot comparison of Mixtral 8x7B.

Ablation Study

Model	Merge	ARC-c	ARC-e	BoolQ	HellaSwag	MMLU	OBQA	RTE	Winogrande	Average
Qwen 60x2.7B	None	0.3951	0.7012	0.8135	0.5932	0.6047	0.310	0.7329	0.6559	0.6008
	Frequency	0.3660	0.6578	0.7948	0.5520	0.5332	0.272	0.7509	0.6464	0.5716
	Average	0.3584	0.6553	0.7936	0.5516	0.5348	0.270	0.7473	0.6559	0.5709
	Fix-Dom	0.3695	0.6692	0.7896	0.5555	0.5338	0.262	0.7365	0.6535	0.5712
	Avg	0.3646	0.6608	0.7927	0.5530	0.5339	0.268	0.7449	0.6519	0.5712
Qwen 45x2.7B	Frequency	0.3532	0.6149	0.7535	0.4695	0.4534	0.228	0.6606	0.6456	0.5223
	Average	0.3575	0.6145	0.7554	0.4706	0.4531	0.228	0.6643	0.6488	0.5240
	Fix-Dom	0.3439	0.6132	0.7544	0.4679	0.4445	0.228	0.6643	0.6504	0.5208
	Avg	0.3515	0.6142	0.7544	0.4693	0.4503	0.228	0.6631	0.6483	0.5224

Table 4. Various merging methods with HC average linkage based on expert outputs.

Model	Calib-Dataset	ARC-c	ARC-e	BoolQ	HellaSwag	MMLU	OBQA	RTE	Winogrande	Average
Qwen 60x2.7B	-	0.3933	0.6982	0.8119	0.5938	0.6034	0.314	0.7401	0.6606	0.6029
	C4	0.3660	0.6578	0.7948	0.5520	0.5332	0.272	0.7509	0.6464	0.5716
	MATH	0.3746	0.6662	0.7917	0.5478	0.5293	0.284	0.7437	0.6425	0.5725
	CodeQA	0.3490	0.6578	0.7976	0.5275	0.5194	0.260	0.7581	0.6448	0.5643
	Avg	0.3532	0.6149	0.7535	0.4695	0.4534	0.228	0.6606	0.6456	0.5223
Qwen 45x2.7B	C4	0.2978	0.5513	0.7593	0.4151	0.4415	0.244	0.6823	0.6164	0.5010
	MATH	0.3089	0.5720	0.7633	0.4179	0.4412	0.214	0.6606	0.6069	0.4981

Table 5. Ablation study on choice of calibration dataset of eight zero-shot language tasks on Qwen.

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Questions?



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